

Probability and Statistics for Biologists

DATA405 Genomic Data Science · Lecture 11 · Sep 29, 2026

ATGCCGGAGGCCCCGCCTCTGCTGTTGGCAGCTGTGTTGCTGGGCCTGGTGCTGCTGGTGGTG

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From a coin to a ChIP-seq peak, DESeq2, and reads you cannot see

Two words, two directions

Probability: given the process, what data do we expect?

Statistics: given the data, what was the process?

process $\longrightarrow$ data | data $\longrightarrow$ process

Two directions, one coin

Probability — the process is known: a fair coin.

Predict the data: A coin you know is fair → you predict about 50 heads in 100 flips, and you can say how often 59 or more would happen (about 9 %).

Statistics — the data are known: 100 flips, 59 heads.

Infer the process: You flipped a coin 100 times and got 59 heads → is this a fair coin, or a loaded one, and if loaded, how much? You cannot see the coin's bias; you infer it.

Part 1

Probability

What a probability is, formally — on a coin and on DNA.

Sample space and events

Sample space Ω = all possible outcomes. **Event** = a subset of Ω .

A coin flip: $\Omega = \{H, T\}$

events: 'heads' = $\{H\}$ 'tails' = $\{T\}$

A genomic position: $\Omega = \{A, C, G, T\}$

events: 'A' = $\{A\}$ 'purine' = $\{A, G\}$ 'pyrimidine' = $\{C, T\}$ 'not A' = $\{C, G, T\}$

Rule 1 — a probability is a number between 0 and 1

0 = impossible. **1** = certain.

$$0 \leq P(A) \leq 1$$

Rule 2 — the outcomes add up to 1

Something must happen.

At a genomic position: four possible events {A}, {C}, {G}, {T}, each with its probability.

$$P(\{A\}) + P(\{C\}) + P(\{G\}) + P(\{T\}) = 0.265 + 0.234 + 0.236 + 0.265 = 1$$

$$\sum \text{ over all outcomes } P(\text{outcome}) = 1$$

Rule 3 — exclusive events add

At one position you cannot see both A and G: the events {A} and {G} are mutually exclusive.

So $P(\text{purine}) = P(\{A\} \text{ or } \{G\}) = P(\{A\}) + P(\{G\}) = 0.265 + 0.236 = 0.50$

$$P(A \text{ or } B) = P(A) + P(B) \quad \text{if } A \text{ and } B \text{ are mutually exclusive}$$

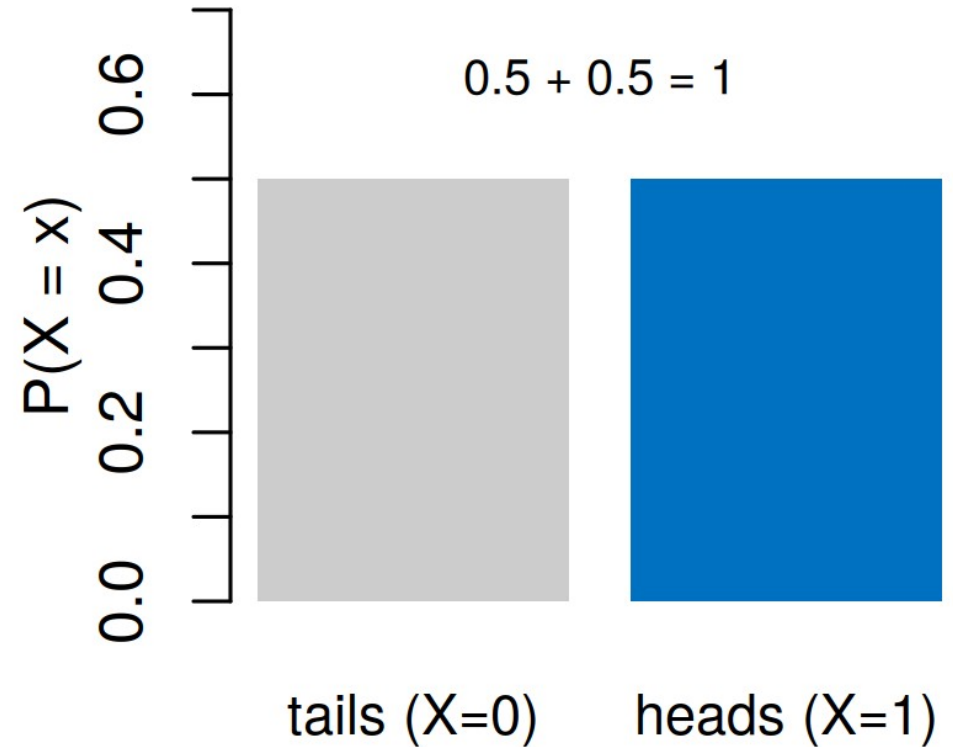
Random variable

To study probability mathematically, events must become numbers.

A random variable turns each outcome into a number.

$X = 1$ if heads, 0 if tails. Y = number of reads in a window.

Its distribution lists $P(X = \text{each value})$ — and those obey rules 1–3.



The distribution of X for one fair flip

Probability's direction

Know the process $\rightarrow$ compute the data.

Fair coin: $P(H) = 0.5$, $P(HH) = 0.25$, $P(HHH) = 0.125$

model $\rightarrow$ data

Part 2

Statistics, with a coin

Given the data, what is the process? A model, a parameter, an estimate, a p-value, an interval.

Statistics

From data, infer the process that produced it.

First write down a model: a family of processes, indexed by a parameter, and then use the data to infer the parameters

$$\text{data} \longrightarrow \text{model}(\text{parameter})$$

The coin model

Each flip lands heads with probability p ; flips are independent.

p is a parameter: a fixed, unknown property of the coin.

'Is the coin loaded?' becomes 'is $p = 0.5$?'

$$X = \text{number of heads in } n \text{ flips} \sim \text{Binomial}(n, p)$$

The data

Data: 100 flips, 59 heads.

Infer p : $\hat{p} = 59/100 = 0.59$

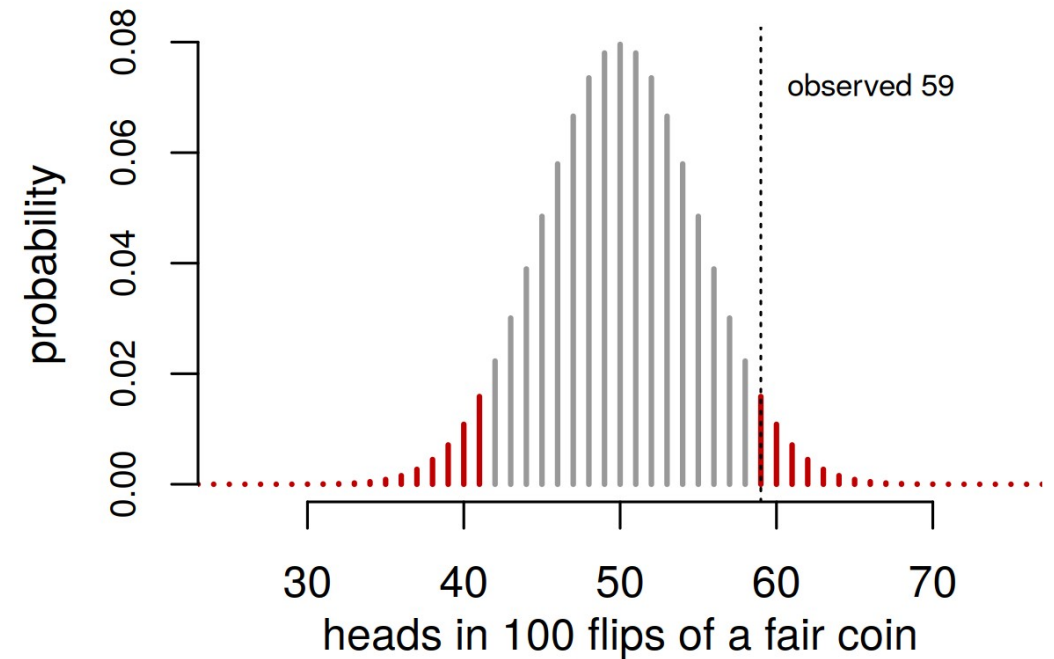
hat = computed from data, changes every sample. no hat = fixed, unknown.

Is 0.59 evidence of loading? Ask the fair coin.

If $p = 0.5$, how often would 100 flips land at least this far from 50?

(≥ 59 or ≤ 41 heads) $\rightarrow$ 8.9 %

That number is the p-value.



$\text{Binomial}(100, 0.5)$; red = results at least as extreme as 59

Two views of the same evidence

p-value: how often a fair coin gives ≥ 59 or ≤ 41 heads $\rightarrow 0.089$

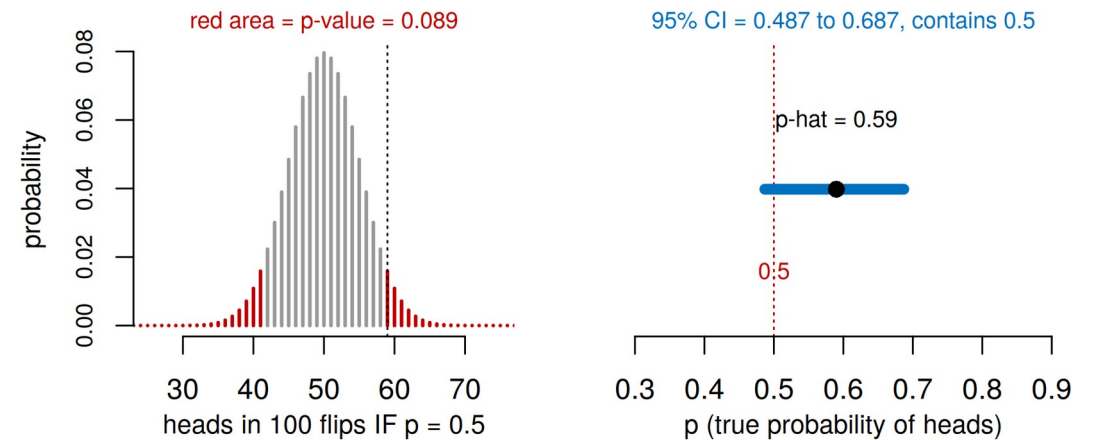
To make a decision, we need a p-value cutoff:
For a single test, the usual cutoff is $p < 0.05$.

95 % confidence interval:

Note $1 - 0.95 = 0.05$

The CI of 59/100 $\rightarrow 0.487 - 0.687$

The interval contains 0.5 $\iff p > 0.05$



Left: the null distribution and the p-value. Right: the interval for p.

More flips, stronger evidence (same 59 % heads)

n	heads	p-value	95 % CI for p
10	6	0.75	0.26 – 0.88
100	59	0.089	0.49 – 0.69
1,000	590	1×10^{-8}	0.56 – 0.62
10,000	5,900	1×10^{-72}	0.58 – 0.60

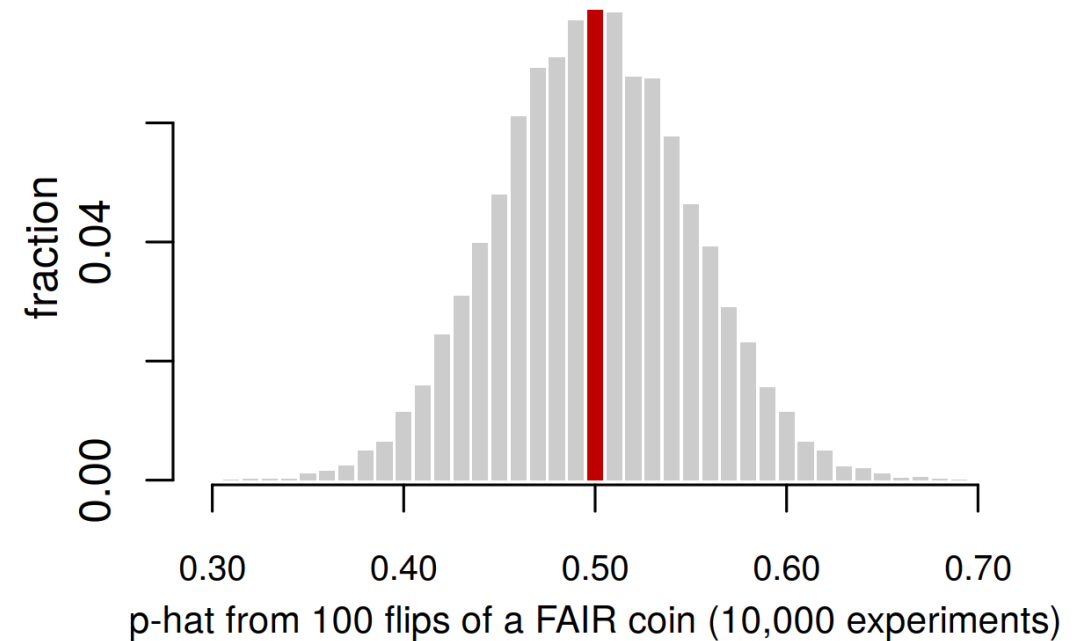
The p-value measures evidence, not the size of the loading.

You never land exactly on 0.5

10,000 experiments, 100 flips each, a truly fair coin:

$\hat{p} = 0.50$ exactly in only 7.9 % of them.

68 % fall between 0.45 and 0.55.



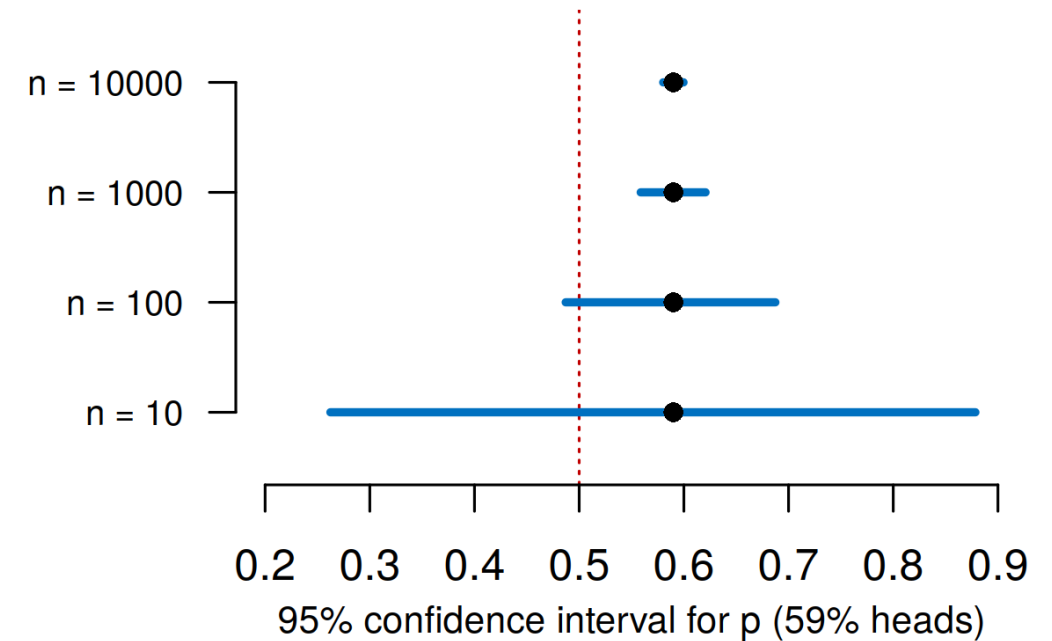
Sampling distribution of $\hat{p}$ for a fair coin

Therefore: an interval

The 95 % confidence interval = the set of p values compatible with the data.

59 / 100: 0.487 – 0.687

In the long run, 95 % of intervals built this way contain the true p.



CI narrows as n grows; 0.5 leaves it at n = 1,000

Where the numbers come from — one line

R

```
binom.test(59, 100, 0.5)

# number of successes = 59, number of trials = 100,
# p-value = 0.08863
# alternative hypothesis: true probability is not equal to 0.5
# 95 percent confidence interval:
#  0.4871  0.6874
# sample estimates:
# probability of success  0.59
```

data → p-value → interval → estimate. Yesterday's t.test prints the same four things.

Interval and test are one calculation

CI contains 0.5 $\iff p > 0.05$

Report the interval; the test is its yes/no summary.

estimate $\rightarrow$ interval $\rightarrow$ decision

Part 3

The same model in ChIP-seq

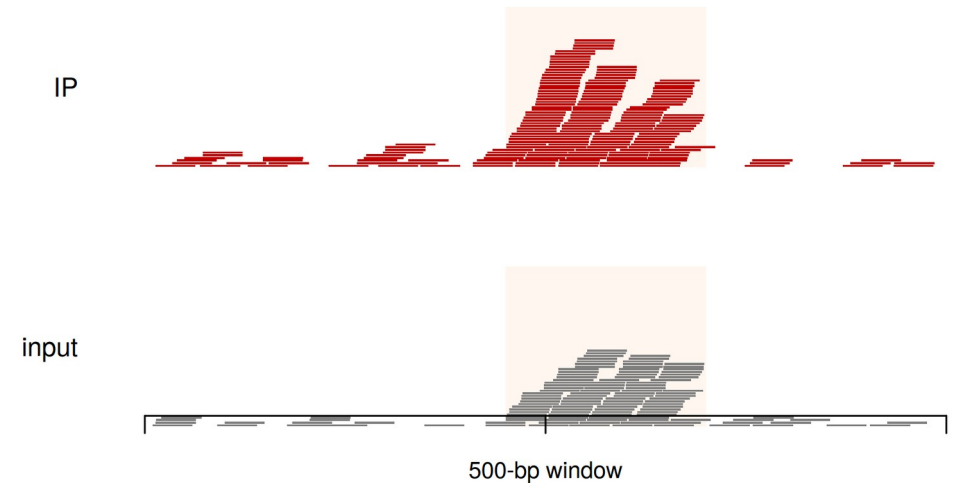
A transcription-factor binding site is a loaded coin.

One window, two piles of reads

IP: DNA pulled down by the antibody.

Input: the same chromatin, no antibody.

Is the tall pile evidence of binding — or chance?



Reads over a 500-bp window (shaded) in IP and input

What produces the pile

The count of reads in a window is a random variable.

Its natural model is Poisson (MACS uses it).

For simplicity we use the binomial — it asks the same question.

First, make the libraries comparable

IP: 20 million reads. Input: 40 million reads.

Input piles are twice as tall for no biological reason.

Normalize to equal depth: input count $\times$ 20/40.

Window A: IP 40, input 20 $\rightarrow$ 40 vs 10

The coin in the window

50 reads in window A. Each came from IP or from input.

No binding: a read is equally likely from either library → a fair coin.

Binding: IP is enriched → a loaded coin.

The model

IP reads $\sim$ Binomial(IP + input, θ)

Null: $\theta = 0.5$ (no binding). Alternative: $\theta > 0.5$.

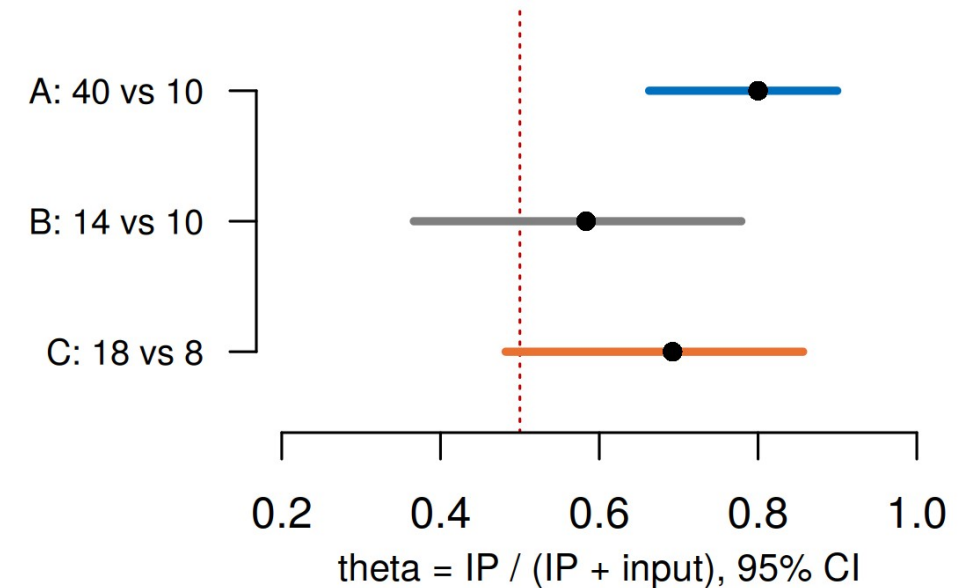
$$\theta = \text{IP} / (\text{IP} + \text{input})$$

Region A — succeeds

IP 40, input 10.

$\hat{\theta} = 0.80$ 95 % CI 0.66 – 0.90 $p = 2.4 \times 10^{-5}$

The interval excludes 0.5. Beyond reasonable doubt: a peak.



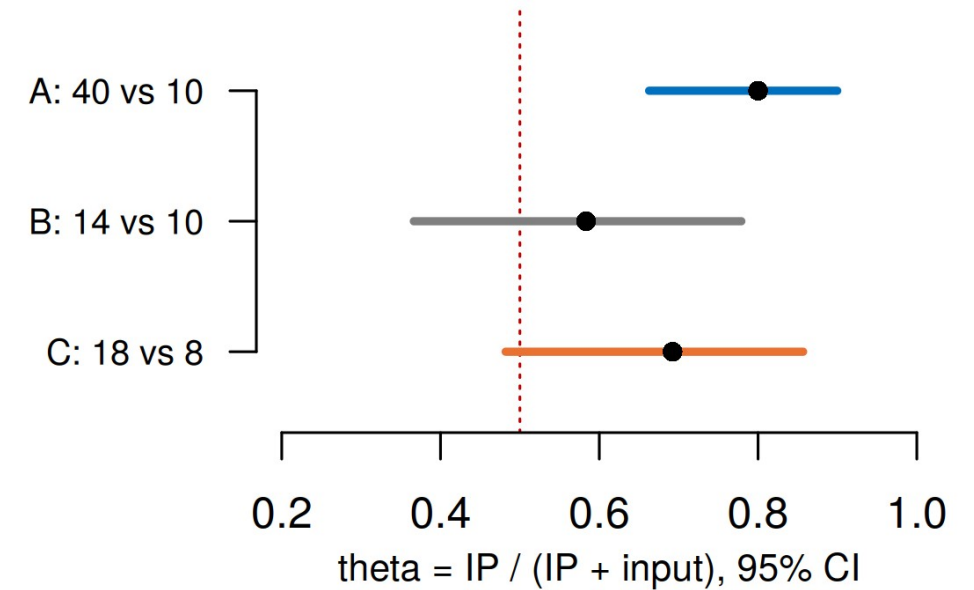
Three regions: estimate and 95 % CI; dashed line = no binding

Region B — fails

IP 14, input 10.

$\hat{\theta} = 0.58$ 95 % CI 0.37 – 0.78 $p = 0.54$

The interval includes 0.5. Not guilty.



Middle bar: Region B

Both from the same line

R

```
binom.test(40, 50, 0.5)      # Region A
#   p-value = 2.4e-05
#   95 percent confidence interval: 0.66 0.90
#   probability of success 0.80

binom.test(14, 24, 0.5)      # Region B
#   p-value = 0.54
#   95 percent confidence interval: 0.37 0.78
#   probability of success 0.58
```

Same call as the coin. Only the names changed.

Region C

YOUR TURN IP 18, input 8. Peak or not? Predict, then read the interval.

- A. Peak — $\hat{\theta} = 0.69$ is clearly above 0.5
- B. Not a peak — need to see the p-value first
- C. Cannot tell without more reads

ANSWER $\hat{\theta} = 0.69$, CI 0.48 – 0.86, $p = 0.076$. Not guilty at $\alpha = 0.05$ — but the interval barely touches 0.5. More reads would settle it (the n table of Part 2).

One region done. Now the genome.

chr22 alone: ~100,000 windows. Whole genome: ~6 million.

We are about to run the same court case 100,000 times.

Part 4

Thousands of regions

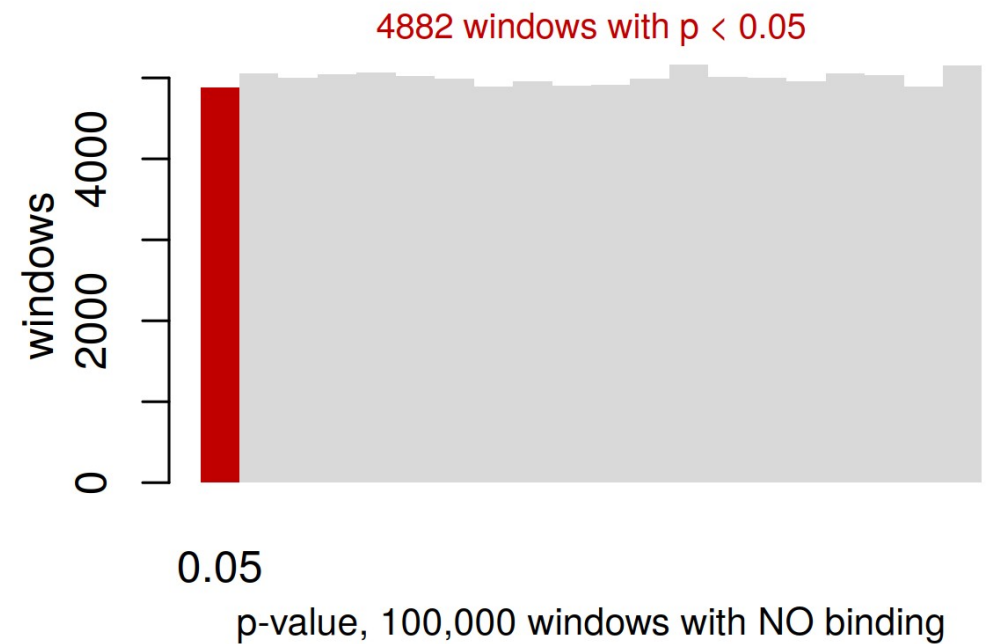
A court that hears 100,000 cases convicts thousands of innocents.

No binding anywhere — still 5,000 peaks

Simulate 100,000 windows with $\theta = 0.5$ everywhere.

4,882 of them have $p < 0.05$.

$\alpha = 0.05$ means: 5 % of true nulls get convicted.



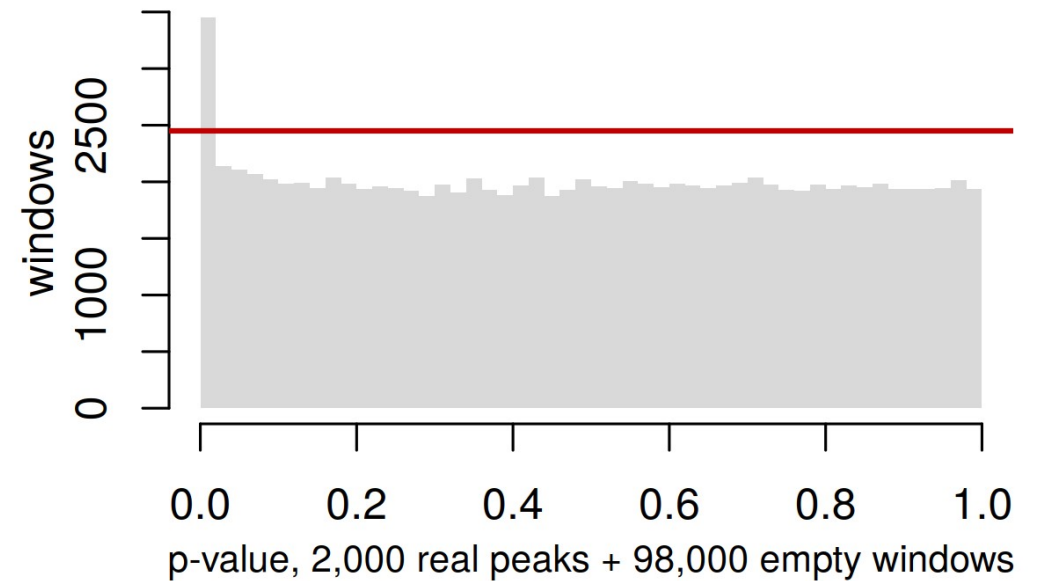
p-values under the null are uniform; the red slice is 5 %

With real peaks mixed in

2,000 real peaks + 98,000 empty windows.

$p < 0.05$ calls 6,701 — of which 4,875 are false.

A list that is 73 % wrong.



Spike near 0 = real peaks; the flat floor = false positives

The simplest fix: demand a much smaller p

Bonferroni: α / number of tests = $0.05 / 100,000 = 5 \times 10^{-7}$

Simulation: 76 called, 0 false.

Safe — and it discarded 96 % of the real peaks. Region A ($p = 2.4 \times 10^{-5}$) would not survive.

A different goal

For a list of peaks we can live with some false ones —
as long as we know what fraction.

That fraction is the false discovery rate (FDR).

The q-value idea

Sort all p-values, smallest first.

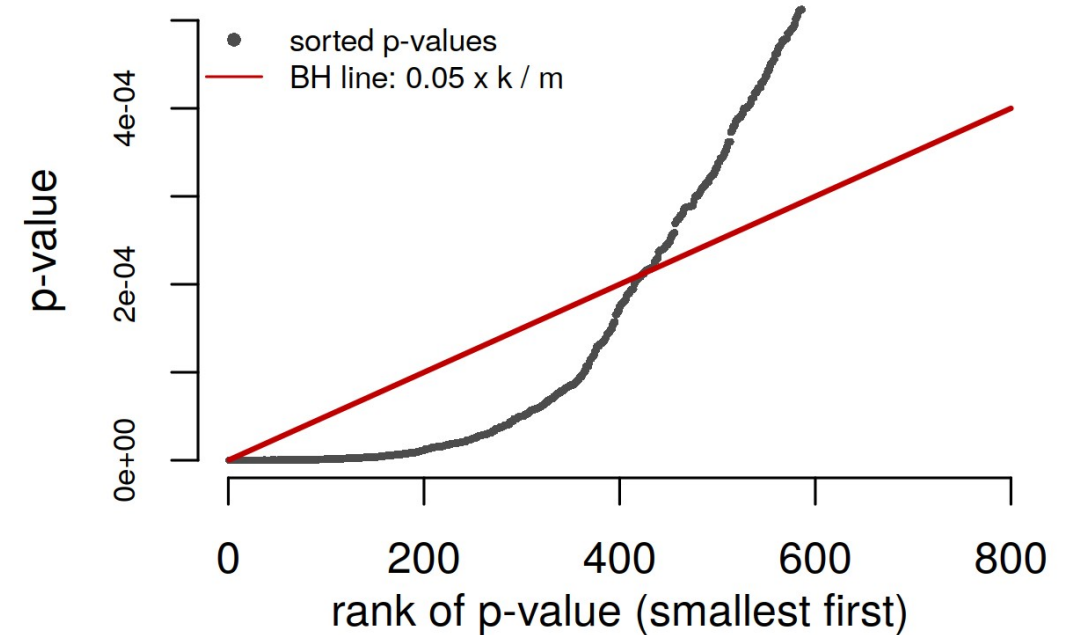
Draw the line at any region. Ask: if I call everything up to here, what fraction of the list is expected to be false?

That fraction is the region's q-value.

Benjamini-Hochberg computes it (1995)

k-th smallest of m p-values: $q \approx p \times m / k$

Keep everything below the last point
under the line $0.05 \cdot k / m$.



Simulated genome: sorted p-values vs the BH line

On the simulation

$p < 0.05$: 6,701 called, 4,875 false

Bonferroni: 76 called, 0 false

$q < 0.05$: 425 called, 15 false (4 %)

FDR is the working compromise of genomics.

One line of R

R

```
q <- p.adjust(p, method = "BH")  
sum(q < 0.05)           # → 425
```

```
# DESeq2's padj column and MACS's -log10(qvalue) are this number
```

Filter and rank by q, never by p, whenever you ran more than one test.

Reading a q-value

$q = 0.08$ for a region means:

a list cut at this region is expected to contain about 8 % false peaks.

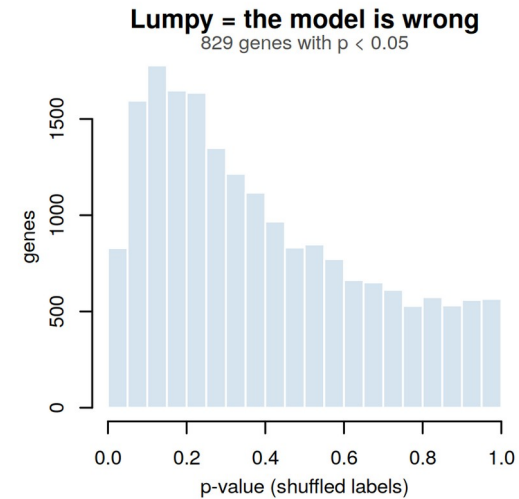
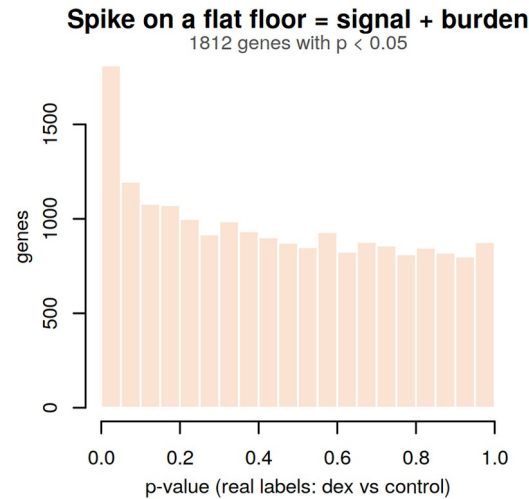
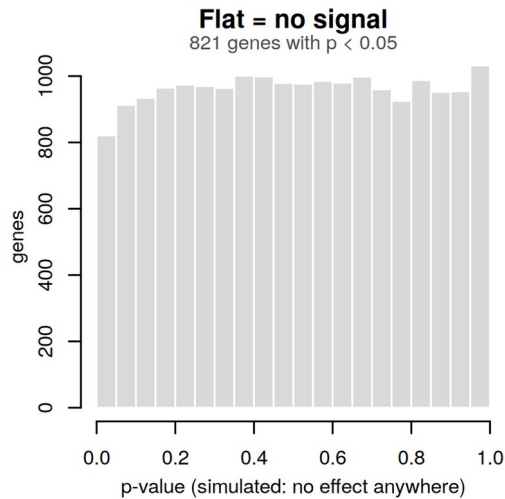
It does not mean an 8 % chance that this region is false.

Always look at the histogram first

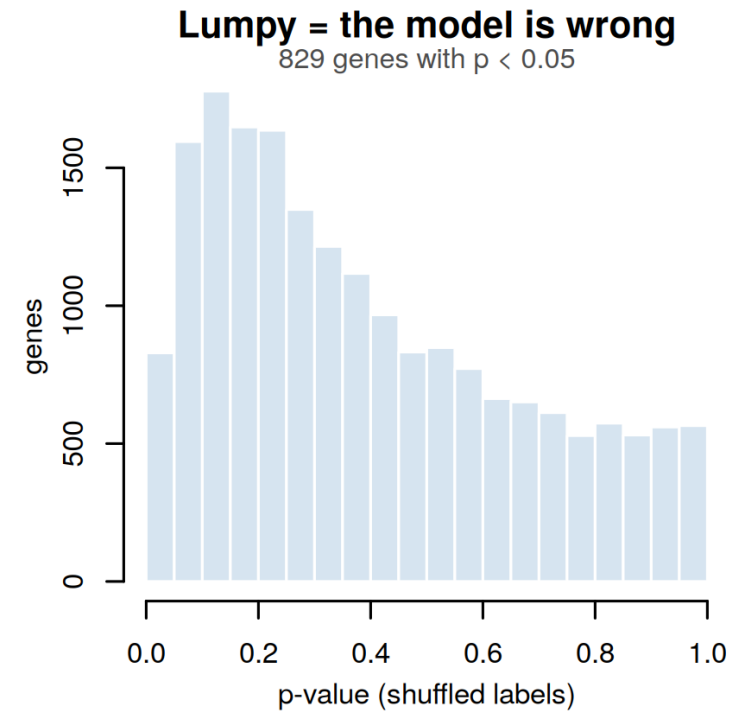
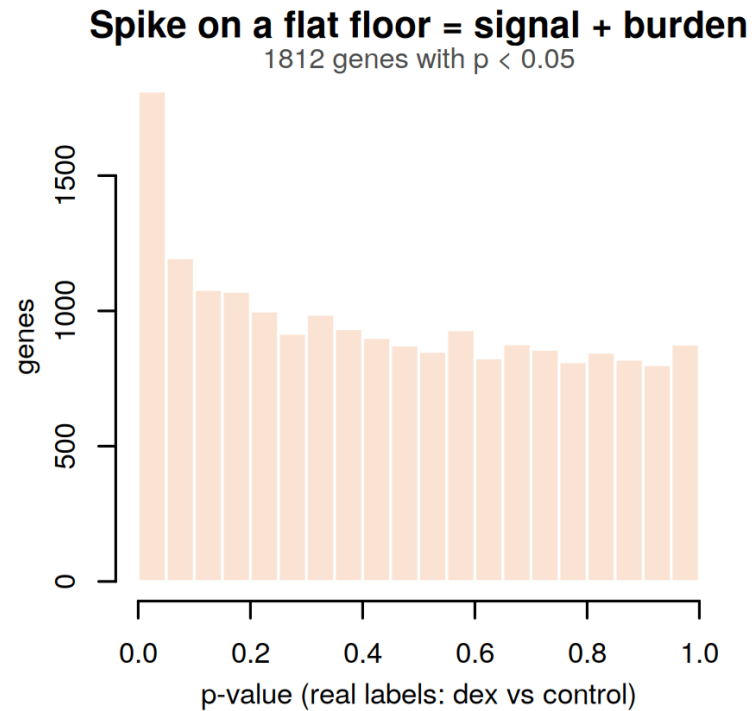
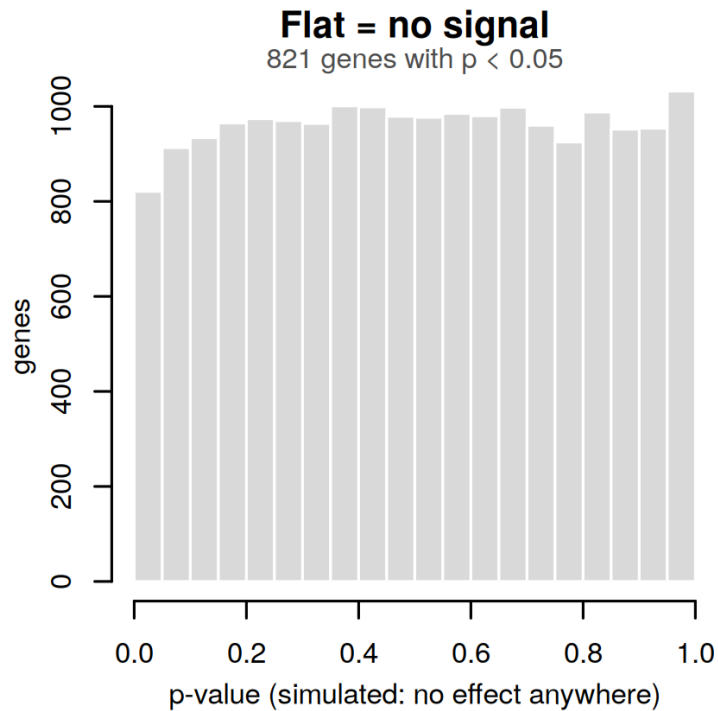
Flat = no signal.

Spike near 0 on a flat floor = signal plus burden.

Sloping up toward 1, or lumpy = the model is wrong.



Three histograms, three verdicts



Left: simulated, no effect. Middle: airway, real labels. Right: airway, shuffled labels.

Flat = no signal. Spike on a flat floor = signal + burden. Lumpy or sloping = the model is wrong.

Part 5

Estimation with hidden data

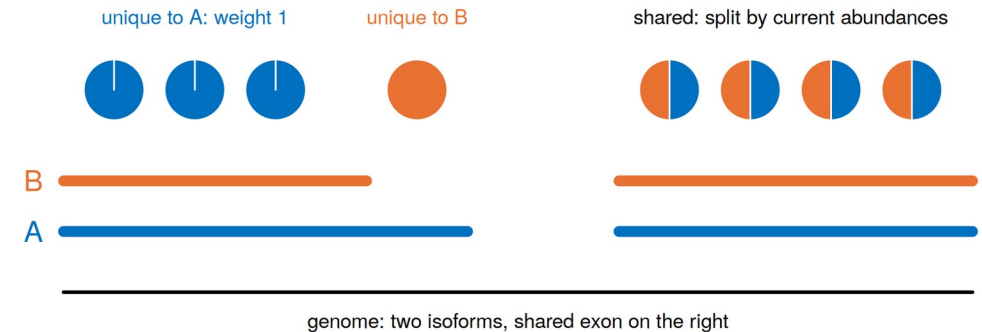
When you cannot see which coin you flipped — EM, with the RNA-seq example step by step.

The problem in RNA-seq

A gene with two isoforms that share an exon.

A read from the shared exon fits both.

To estimate abundances you must know where each read came from; to assign a read you must know the abundances.



Reads unique to A, unique to B, and shared

The two-jar coin version

Jar A coins: heads 70 %. Jar B coins: heads 30 %.

Records of 10 flips each — but nobody wrote down the jar.

Estimate the two biases.

Break the circle by alternating

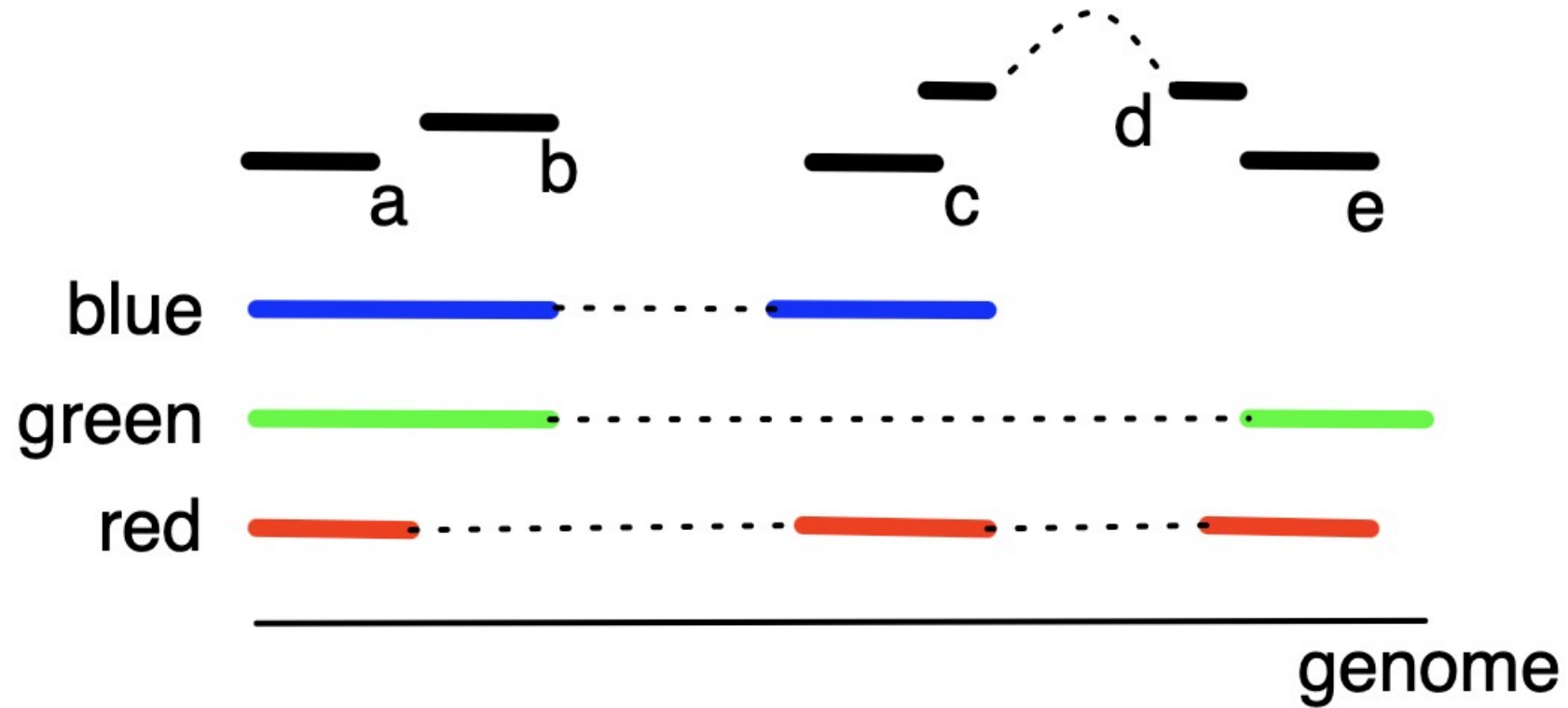
Guess the parameters.

E-step: assign hidden labels in expectation (fractionally).

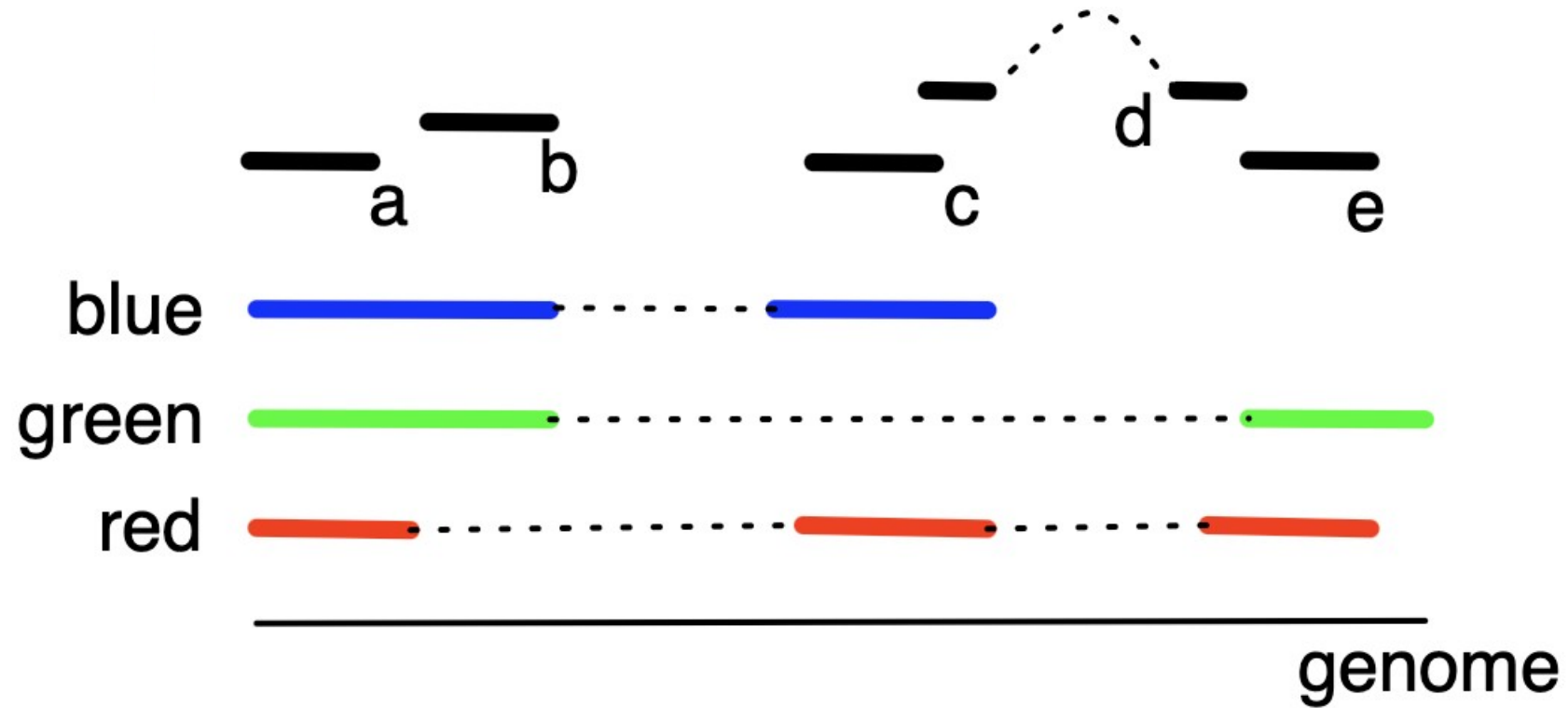
M-step: re-estimate the parameters from the assignments.

Repeat until nothing changes.

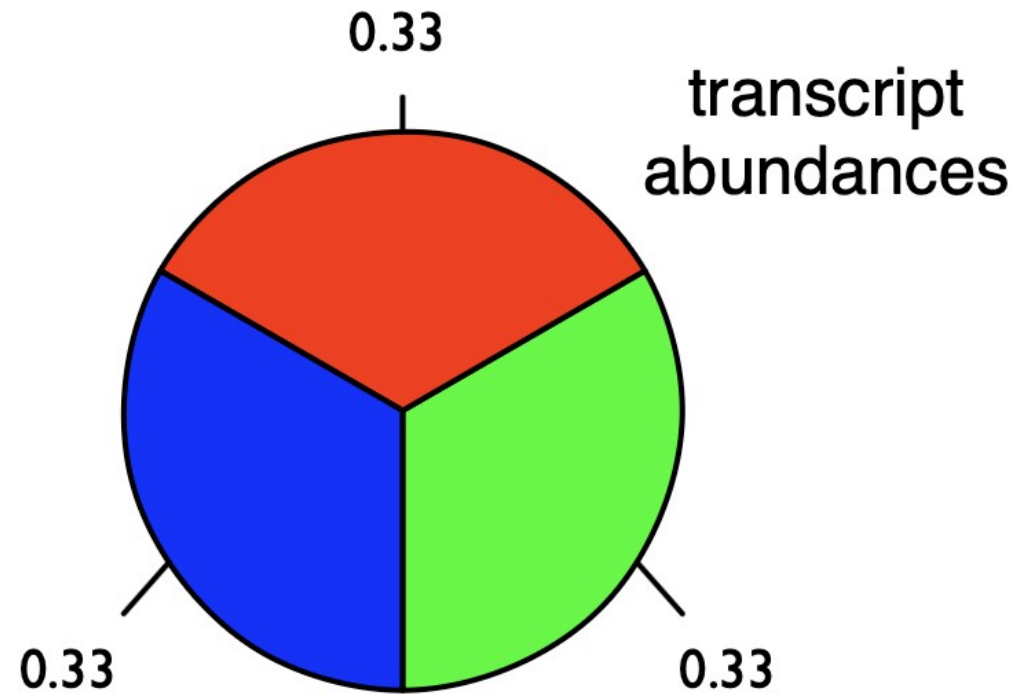
Let's learn by example and use just you intuition!



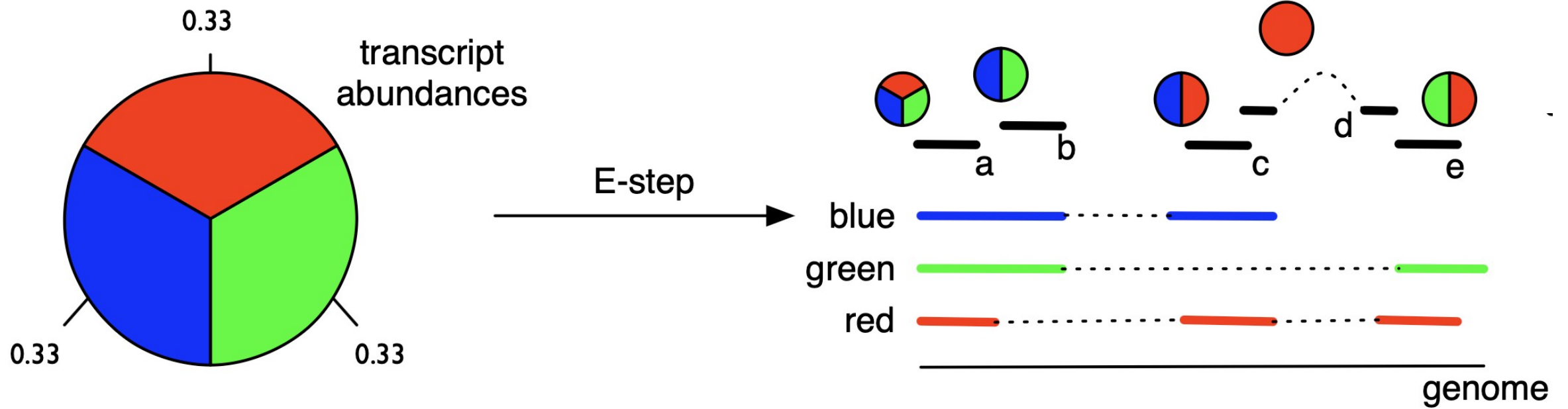
E-M Principle: reads are more likely to be mapped to transcripts with higher abundances



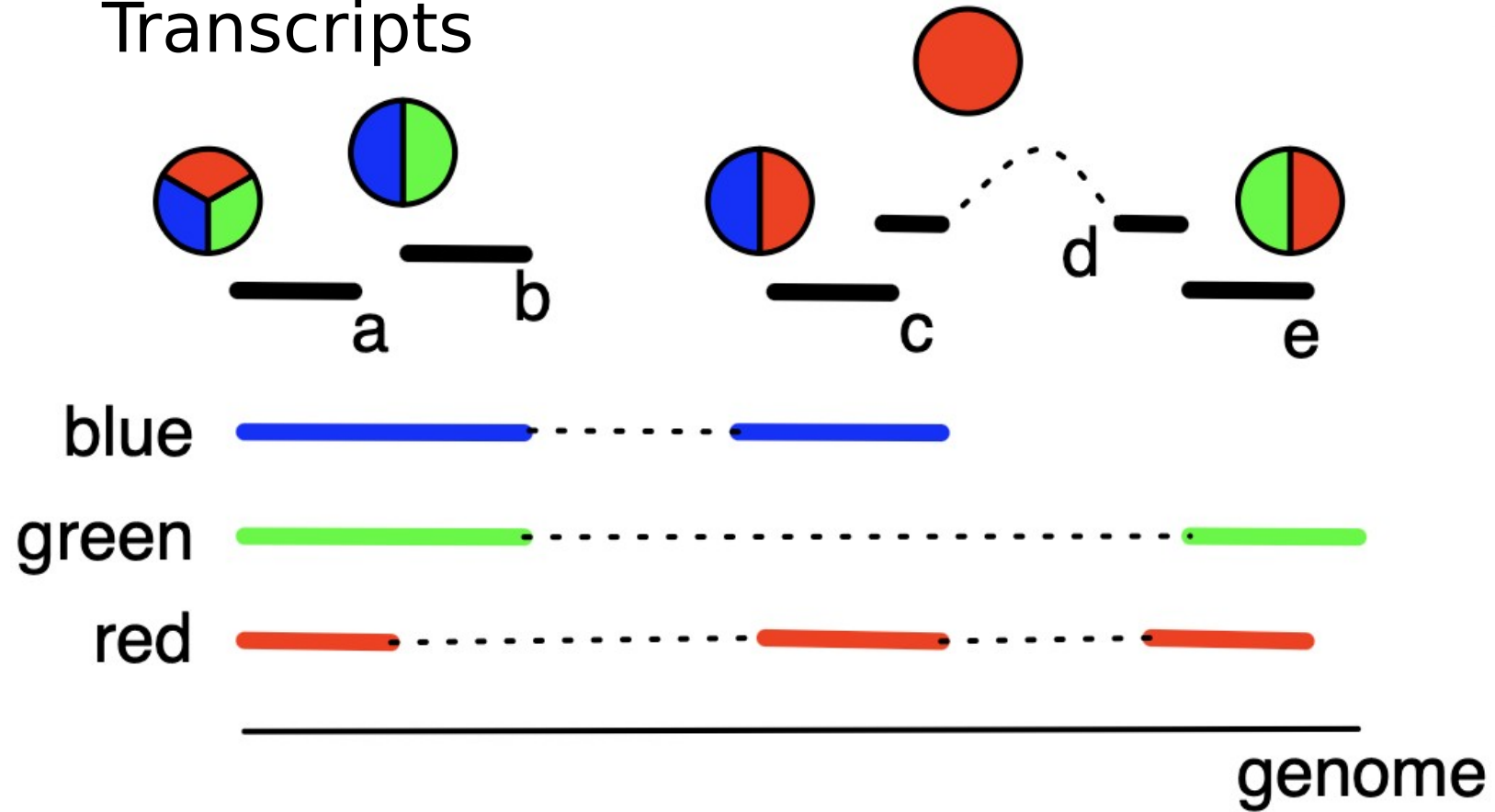
Initiation: All the transcripts have the same abundances



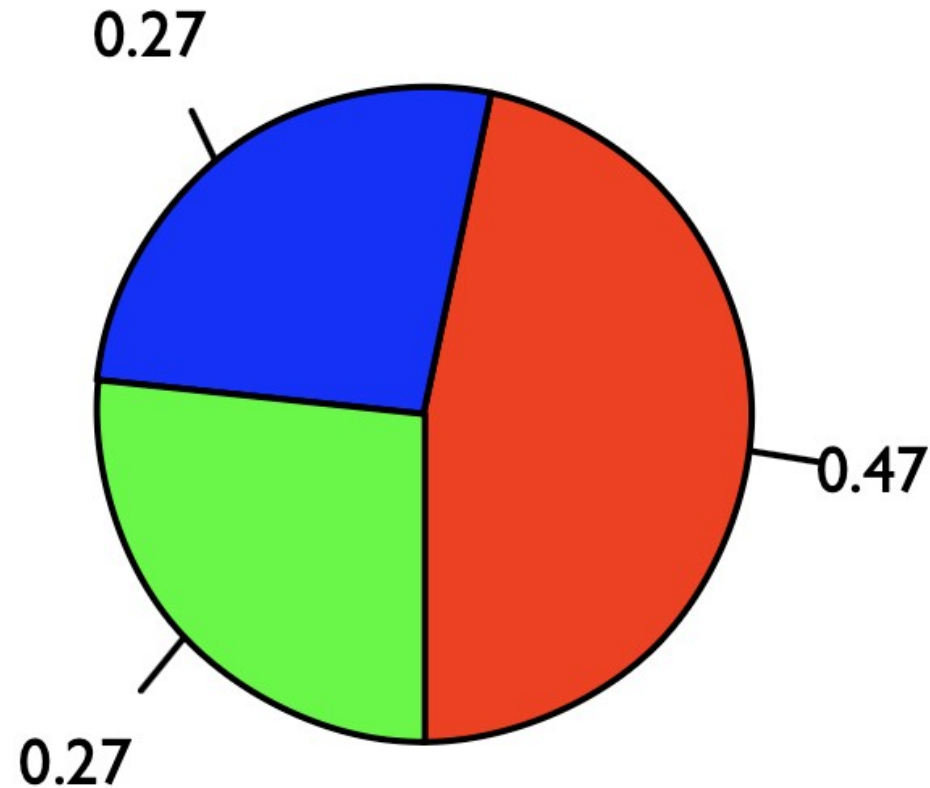
E-Step: Update the Classification of Reads



M-Step: Estimate the Abundances of Transcripts

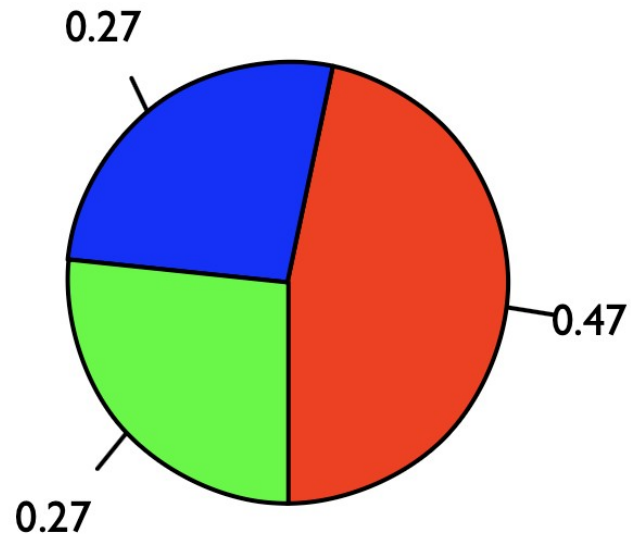


M-Step results in an **Updated Transcript Abundances**

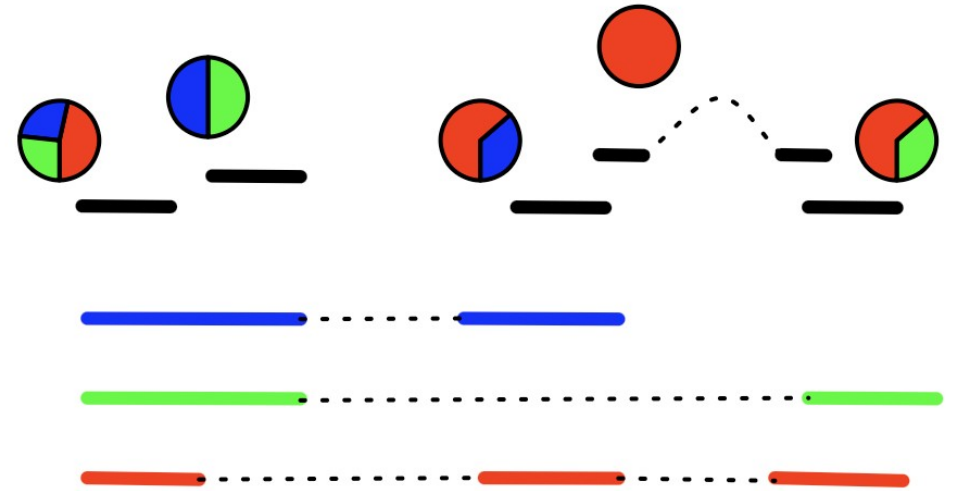


E-Step: Update the Classification of

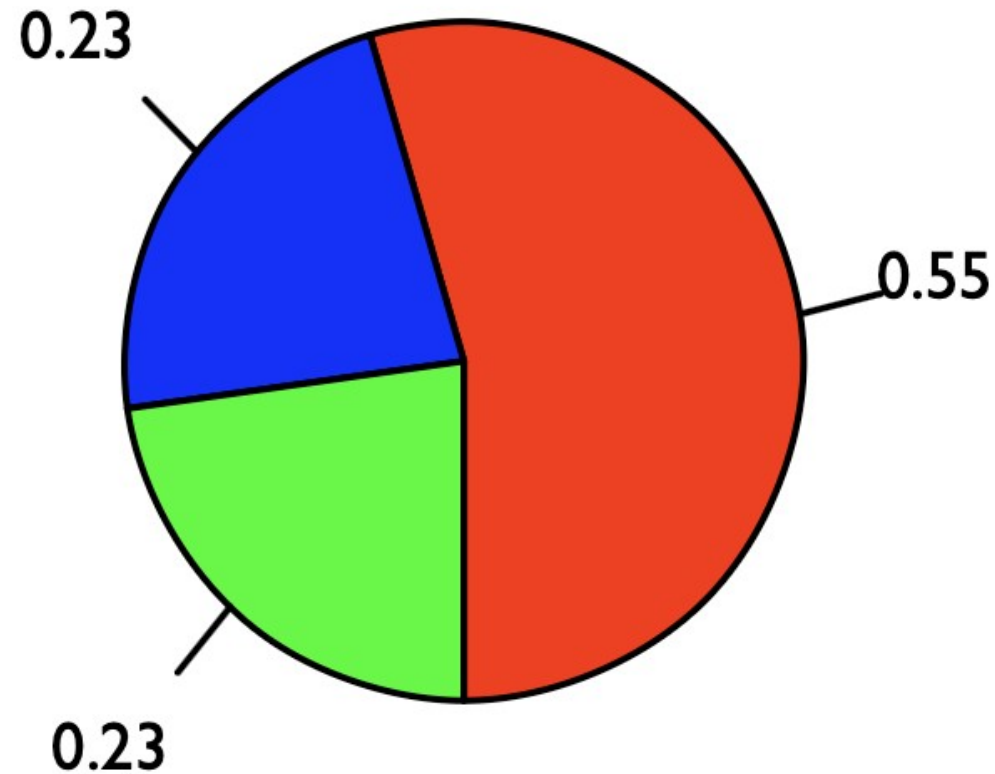
Nodes

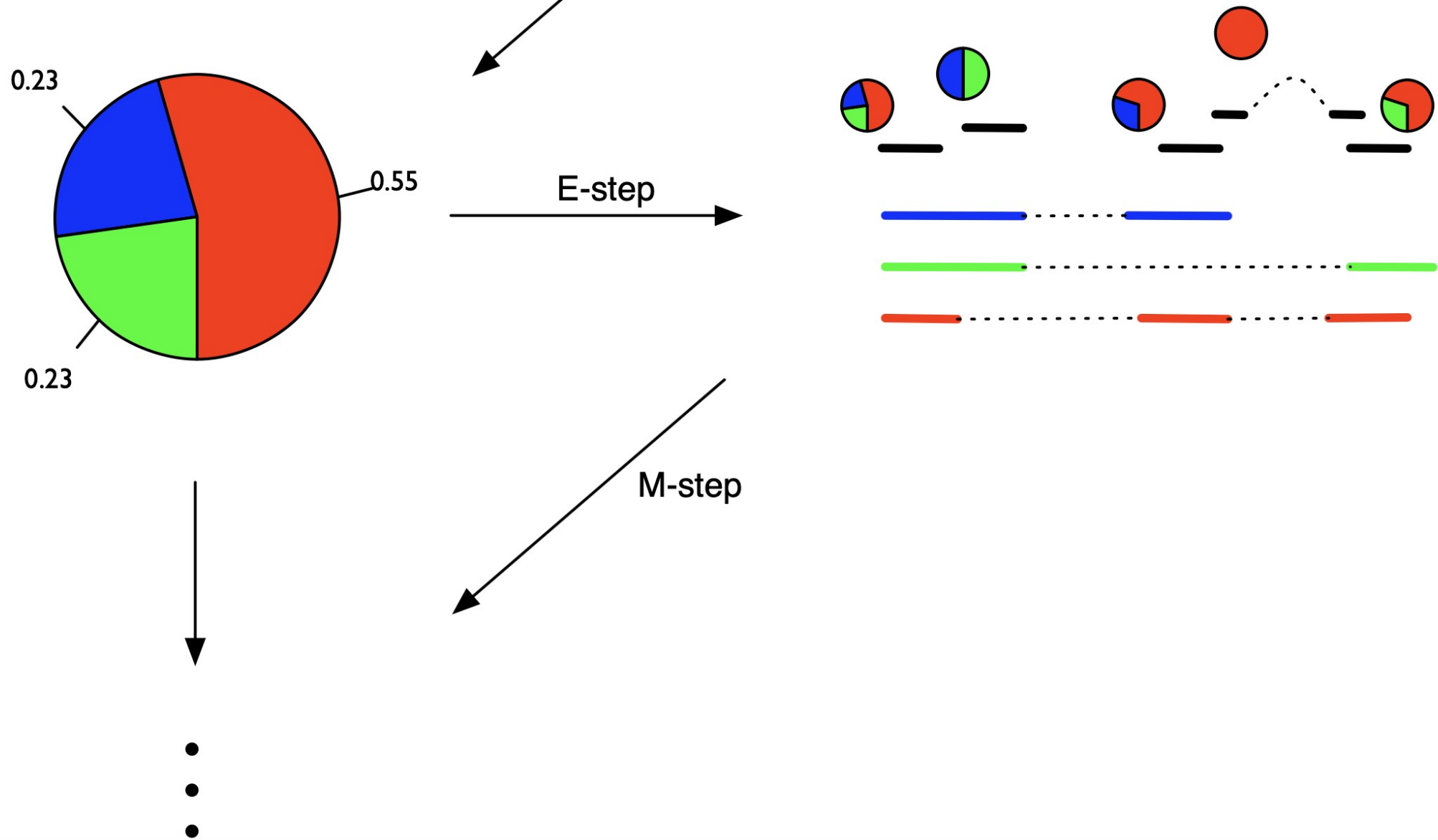


E-step



M-Step results in an **Updated Transcript Abundances**





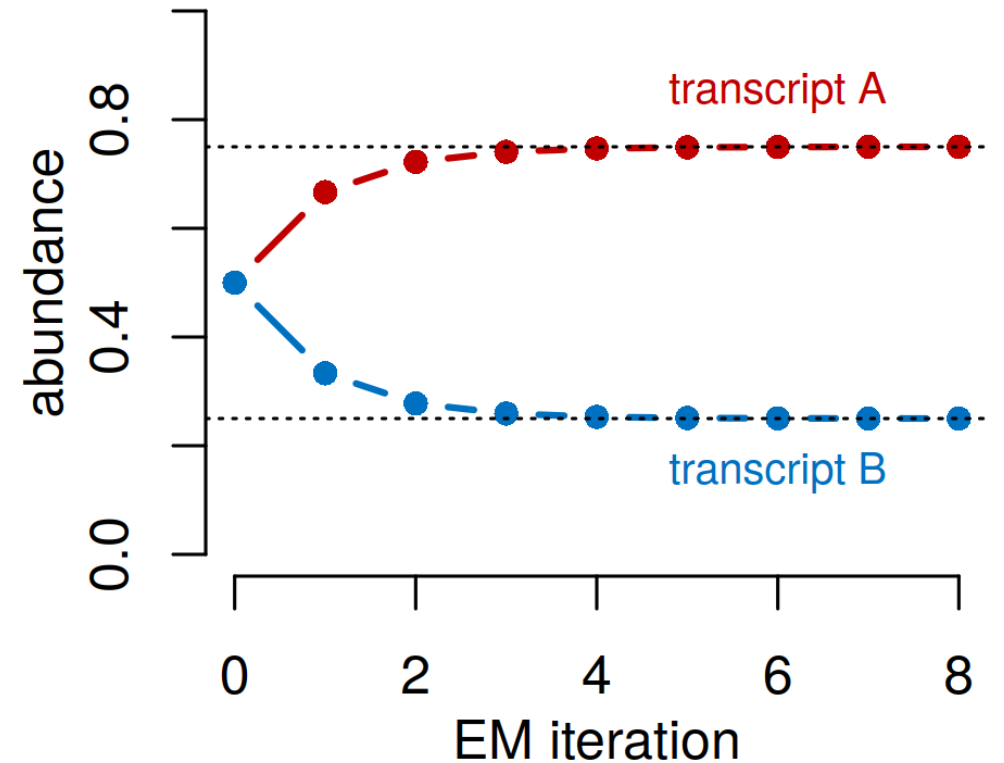
Do E-Step and M-Step iteratively, until transcript abundances converge

A worked example

6 reads unique to A, 2 unique to B, 4 shared.

$0.50 \rightarrow 0.667 \rightarrow 0.722 \rightarrow 0.741 \rightarrow 0.747 \rightarrow 0.75$

Naive half-splitting says 0.667.



Abundance of A (red) and B (blue) over EM iterations

Ten lines

Python

```
nA, nB, nShared = 6, 2, 4
theta = [0.5, 0.5]
for it in range(8):
    eA = nA + nShared * theta[0] / sum(theta)    #
    E-step
    eB = nB + nShared * theta[1] / sum(theta)
    theta = [eA / (eA + eB), eB / (eA + eB)]    #
    M-step
    print(it + 1, [round(t, 3) for t in theta])
# 1 [0.667, 0.333] ... 6 [0.75, 0.25]
```

R

```
nA <- 6; nB <- 2; nShared <- 4
theta <- c(0.5, 0.5)
for (it in 1:8) {
    eA <- nA + nShared * theta[1] / sum(theta)    # E-
    step
    eB <- nB + nShared * theta[2] / sum(theta)
    theta <- c(eA, eB) / (eA + eB)                # M-
    step
    cat(it, round(theta, 3), "\n")
}
# 1 0.667 0.333 ... 6 0.75 0.25
```

Runs on the course site in both languages.

Why the answer is right

Each round can only increase the likelihood of the data.

The fixed point is self-consistent: the abundant transcript claims more of the ambiguous reads, which makes it more abundant.

RSEM output is the last E-step

expected_count = 317.84 for IGLC2 on chr22 — fractional, because reads were split by weight.

ACTBP15, a pseudogene: 650 reads — because only chr22 was in the reference.

EM chooses among the options it is given. Give it the whole genome.

One table

Question	Model	Parameter	Tool
allele at a site	Binomial	θ	binom.test
IP vs input in a window	Binomial (Poisson in MACS)	$\theta = \text{IP}/(\text{IP} + \text{input})$	binom.test → MACS
reads across isoforms	mixture of transcripts	$\theta_A, \theta_B, \dots$	EM (RSEM)
thousands of tests	—	—	BH q-value: p.adjust

What you can now do

State the three rules of probability, and tell a parameter from a probability.

Read `binom.test` output; explain a p-value with a court; explain a q-value with a list.

Explain EM with two jars, and follow the E-step and M-step on reads.

Tomorrow's lab: `binom.test` on real reads, the 100,000-tests trap, BH by hand.

Sources

- Holmes S, Huber W. Modern Statistics for Modern Biology. CUP 2019; free at huber.embl.de/msmb — ch. 1, 2, 6, 8.
- Whitlock MC, Schluter D. The Analysis of Biological Data, 3rd ed. — ch. 4–6. • Motulsky H. Intuitive Biostatistics, 4th ed. — ch. 15–17.
- Diez, Çetinkaya-Rundel, Barr. OpenIntro Statistics (free) — ch. 3, 5. • Wasserman L. All of Statistics — ch. 1.
- Love, Huber, Anders. DESeq2. Genome Biol 2014 • Li, Dewey. RSEM. BMC Bioinformatics 2011 • Zhang et al. MACS. Genome Biol 2008.
- Benjamini, Hochberg. J R Stat Soc B 1995 • Storey. J R Stat Soc B 2002 • Dempster, Laird, Rubin. J R Stat Soc B 1977 • Do, Batzoglou. Nat Biotechnol 2008.
- Xu et al. To mock or not. Nucleic Acids Res 2021 • Himes et al. PLoS ONE 2014 (airway data).

Questions?

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Course site: every example runnable in the browser.